

Practical No. 05

Transportation Problems

Aim:

To solve a Transportation Problem using TORA (Total OR Analysis) Software and determine the optimal transportation schedule and minimum transportation cost.

Problem Statement:

A company has **three warehouses** supplying products to **four stores**. The transportation cost per unit from each warehouse to each store is given below.

Warehouse / Store	S1	S2	S3	S4	Supply
W1	6	4	8	7	50
W2	5	7	6	8	40
W3	9	8	5	6	60
Demand	30	40	50	30	150

Find an initial basic feasible solution to the transportation problem using

(i) North West Corner Method

(ii) Least Cost Method

(iii) Vogel's Approximation Method

Theory:

Transportation Problem

A transportation problem is a special type of **Linear Programming Problem (LPP)** in which a homogeneous product is transported from several **sources** to several **destinations** at minimum cost.

The main objective is to determine an optimal transportation plan that minimizes the total transportation cost while satisfying all supply and demand constraints.

Mathematical Formulation

Let:

x_{ij} = quantity transported from source i to destination j

The objective function is:

$$\text{Minimize } Z = \sum_{i=1}^m \sum_{j=1}^n c_{ij} x_{ij}$$

where:

- c_{ij} = transportation cost from source i to destination j
- x_{ij} = quantity transported from source i to destination j

Supply Constraints

For each source:

$$\sum_{j=1}^n x_{ij} = a_i$$

where a_i represents the available supply.

Demand Constraints

For each destination:

$$\sum_{i=1}^m x_{ij} = b_j$$

where b_j represents the required demand.

Also,

$$x_{ij} \geq 0$$

Important Terms

- **Source**

A location from where goods are supplied, such as a **warehouse, factory or production unit**.

- **Destination**

A location where goods are required, such as a **store, customer or distribution centre**.

- **Supply**

The quantity of goods available at each source.

- **Demand**

The quantity of goods required at each destination.

- **Transportation Cost**

The cost of transporting one unit from a particular source to a particular destination.

- **Balanced Transportation Problem**

A transportation problem is balanced when:

$$\text{Total Supply} = \text{Total Demand}$$

If total supply and total demand are unequal, the problem is called an **unbalanced transportation problem**.

Procedure in TORA Software

- **Step 1:** Open **TORA Software**.
- **Step 2:** Select: **Transportation Model** from the appropriate menu.
- **Step 3:** Enter the number of:
 - Sources = **3**
 - Destinations = **4**
- **Step 4:** Enter the transportation cost matrix:

	S1	S2	S3	S4	Supply
W1	6	4	8	7	50
W2	5	7	6	8	40
W3	9	8	5	6	60
Demand	30	40	50	30	

- **Step 5:** Select the required initial solution method, such as North-West Corner Rule, Least Cost Method, or VAM.
- **Step 6:** Run the transportation algorithm in TORA.